NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

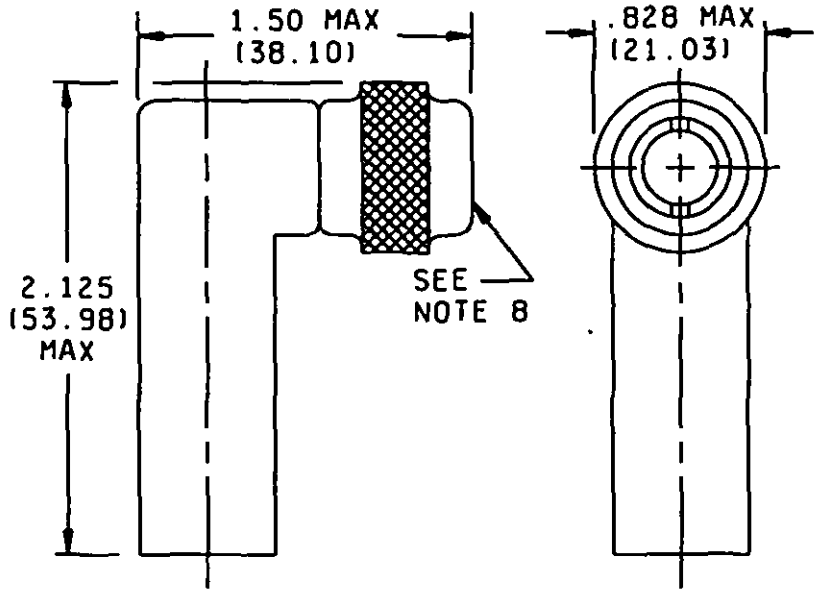
MIL-PRF-39012/84B
3 October 1986
SUPERSEDING
MIL-C-39012/84A
28 July 1983

PERFORMANCE SPECIFICATION

CONNECTORS, PLUGS, ELECTRICAL, COAXIAL, RADIO FREQUENCY, *SERIES QSC (CABLED PIN CONTACT, RIGHT ANGLE, CLASS 2)*

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-PRF-39012.



MARKING IMPLEMENTATION DATE, *CATEGORY B, SEE TABLE III*

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. Dimension 1.500 and .828 are the maximum envelope dimensions.
5. Dimension 2.125 defines the maximum length of the connector when assembled to the appropriate cable.
6. Wrench flats to accommodate standard wrench opening per FED-STD-H28, appendix 10.
7. All undimensioned pictoral configurations are for reference purposes only.
8. Series QSC, pin contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.

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TABLE I. Dash number and usage cross-reference.

Dash number 1/	Applicable cable no. ≡
CATEGORY A - FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED) 4/	
0001 5/	M17/2-RG006Δ M17/112-RG304@ M17/171-00001ø M17/73-RG212* M17/162-00001ø
0002 5/	M17/74-RG213 M17/163-00001ø M17/75-RG214* M17/164-00001ø M17/77-RG216Δ RG225@
0003 5/	M17/78-RG217* M17/165-00001ø
0004 5/	M17/72-RG211* M17/161-00001ø
0005 5/	M17/79-RG218* M17/166-00001ø
CATEGORY C - FIELD REPLACEABLE (MIL-C-22520/5-01 BASIC CRIMP TOOL) 2/	
0013 3/	M17/112-RG304*@≠ M17/171-00001≠ø
0014 3/	M17/173-RG212*≠ M17/162-00001≠ø
0015 3/	M17/74-RG213* M17/163-00001<ø
0016 3/	M17/75-RG214*< M17/164-00001<ø
0017 3/	RG225*@<
0018 3/	M17/6-RG011*Δ<
0019 3/	M17/65-RG165*@< M17/159-00001<ø

See footnotes at end of table.

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1/ Typical mating connectors for each dash number are those connectors of MIL-C-39012/64 which are of the same category and utilize the cable.

2/ Category C connectors are assembled by means of the applicable crimping tool per MIL-C-22520 to the specified cable stripped in accordance with figure 2.

3/ These connectors have captivated center contacts.

4/ Inactive for new design. Not for Air Force or Navy use.

5/ These connectors shall be furnished with noncaptive center contacts.

≡ The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to be used for the +200°C thermal shock tests.

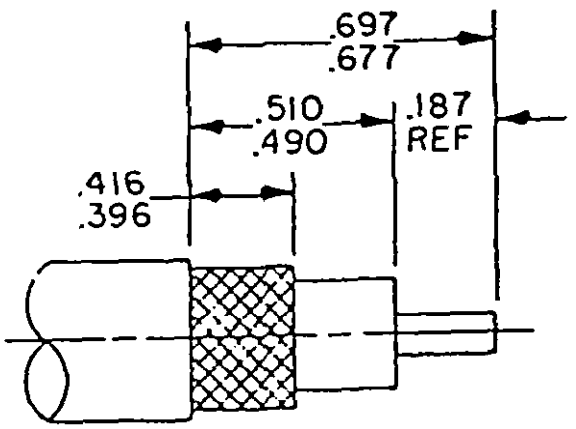
Δ These are not 50 ohm cables, therefore, when attached to the specified connectors, VSWR, RF leakage and insertion loss are not applicable.

≠ M22520/5-35 Closure A or M22520/5-55 Closure A

< M22520/5-61

ø Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.

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Inches	mm
.187	4.75
.396	10.06
.416	10.57
.490	12.45
.510	12.95
.677	17.20
.697	17.70

NOTES:
1. Dimensions are inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Cable stripping dimensions for category C connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

1,500 volts rms maximum working voltage at sea level.

325 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force - Not applicable.

Torque - 2-1/2 inch-pounds maximum.

Coupling proof torque: 15 inch-pounds minimum.

Inspection conditions: 6 to 10 inch-pounds.

Mating characteristics:

See MIL-STD-348.

Outer contact:

Test ring ID - .411 maximum, 16 microinch finish.

Insertion force - 7 pounds maximum when inserted a minimum of .125.

Contacts with slotted members:

Shall contact a .419 minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B. 5,000 megohms minimum.

Center contact retention: 6 pounds minimum axial force. Applicable to captivated-center-contact connectors only.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR):

From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower. 1.30 maximum.

Swept frequency VSWR test setup:

Item 6 - VSWR shall be less than 1.015 + .005 F (F in GHz).

Item 16 - VSWR shall be less than 1.015 + .005 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.045 + .015 F (F in GHz).

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Group B inspection - VSWR shall be less than 1.01 +.01 F (F in GHz).

Group C inspection - VSWR shall not exceed 1.1.

Connector durability:

500 cycles minimum at 12 cycles/minute maximum. The mating force shall not diminish by more than 10 percent.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact:	1.0	1.5
Outer contact:	.20	Not applicable
Braid to body:	.05	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 4,000 volts rms at sea level.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B.

Shock: Method 213 of MIL-STD-202, test condition G.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see table I and II).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Altitude - 70,000 feet.

Voltage - 750 volts minimum.

RF high potential withstanding voltage:

Voltage and frequency - 2,500 volts rms at 5 MHz.

Leakage current - Not applicable.

Cable retention force:

Non-crimp assemblies: 75 pounds minimum.

Crimp assemblies:

50 pounds minimum for cables .155-.189 OD.
60 pounds minimum for cables .190-.229 OD.
75 pounds minimum for cables .230-.249 OD.
90 pounds minimum for cables .250-and larger.

Coupling mechanism retention force: 100 pounds minimum.

RF leakage: -90 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss: 0.15 dB maximum at 10 GHz.

Qualification not required.

Part number: M39012/84 (dash number from table I or "B" number from table II).

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TABLE II. Category B-nonfield replaceable (special tools may be required).

NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY.

M39012/84B 1/ 2/ 3/	Applicable cable ≡
0006	M17/112-RG304*@ M17/171-00001ø
0007	M17/173-RG212* M17/162-00001ø
0008	M17/74-RG213* M17/163-00001ø
0009	M17/75-RG214* M17/164-00001ø
0010	RG225*@
0011	M17/6-RG011*Δ
0012	M17/65-RG165*@ M17/159-00001ø

1/ Typical mating connectors for each dash number are those connectors of MIL-C-39012/64 which are of the same category and utilize the cable.

2/ These connectors shall be furnished with noncaptive center contacts.

3/ Inactive for new design. Not for Air Force or Navy use.

≡ The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to be used for the +200°C thermal shock tests.

Δ These are not 50 ohm cables, therefore, when attached to the specified connectors, VSWR, RF leakage and insertion loss are not applicable.

ø Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.

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TABLE III. Cross-reference of part numbers.

Part number 1/ M39012/84B	Superseded part number M3912/84-
0006	0006
0007	0007
0008	0008
0009	0009
0010	0010
0011	0011
0012	0012

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI, EA
Navy - SH
Air Force - 11, 17, 80, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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